



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024–25
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: MATHEMATICS [SET A]

FULL MARKS: 80
TIME: $2\frac{1}{2}$ HOURS

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

This paper consists of 6 printed pages.

SECTION A

(Attempt all questions from this section)

Question1

Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.) [15]

(i) If $t_n = 5n + 1$ for an AP, then the common difference is

- (a) 1 (b) 5 (c) 6 (d) 11

(ii) Which of the following is invariant when reflected in the line $y + 2 = 0$.

- (a) (10,2) (b) (-10,-2) (c) (-2,-10) (d) (-2,0)

(iii) If a matrix P is of order 1×2 and matrix Q is of order 2×1 , then the matrix QP will contain:

- a) 1 element (b) 2 elements (c) 4 elements (d) QP multiplication is not possible

(iv) If $(x - 2)$ is a factor of $2x^3 - x^2 + px - 2$, then the value of p is

- (a) -11 (b) -5 (c) -2 (d) 5

(v) Which of the following quadratic equation does not have 3 as a root?

- (a) $x^2 - 4x + 3 = 0$ (b) $x^2 + 3x - 18 = 0$
(c) $2x^2 - 7x + 3 = 0$ (d) $3x^2 - 6x - 2 = 0$

(vi) If the scale factor is 1:50000, and the distance between two cities on the map is 2cm, then the actual distance between two cities is:

- (a) 100000km (b) 2500km (c) 1000km (d) 1km

(vii) If $x \in \mathbb{N}$, the solution set of the inequation $5 - 4x \geq 2 - 3x$ is

- (a) $\{0, 1, 2, 3\}$ (b) $\{3, 4, 5, \dots\}$
(c) $\{1, 2, 3\}$ (d) $\{x : x \in \mathbb{R}, x \leq 3\}$

(viii) In a G.P. if $S_{18} - S_{17} = 32$ then

- (a) $a = 32$ (b) $t_{18} = 32$ (c) $t_{17} = 32$ (d) $r = 32$

(ix) If $x, 3, 12$ and y are in continued proportion, then the values of x and y are respectively:

- (a) $4/3$ and 48 (b) $4/3$ and $1/48$ (c) $3/4$ and 48 (d) 1 and 1

(x) Abhinav purchases an article for ₹ 61600 from a dealer, which includes 10% GST. The amount of SGST paid is

- (a) ₹ 2800 (b) ₹ 3580 (c) ₹ 5600 (d) ₹ 6160

(xi) If areas of two similar triangles are 108 cm^2 and 48 cm^2 , then the ratio of their corresponding sides are :

- (a) 9:4 (b) 4:9 (c) 3:2 (d) 2:3

(xii) The centroid of the triangle whose vertices are $(3, 4)$, $(0, 4)$ and $(0, 0)$ is

- (a) $(1.5, 0)$ (b) $(1.5, 8/3)$ (c) $(1.5, 1.5)$ (d) $(1, 8/3)$

(xiii) The inclination of the line $y - 4 = 0$ is

- (a) 0° (b) 45° (c) 60° (d) 30°

(xiv) Anoushka buys x number of shares at par of a certain company paying $y\%$ dividend. The rate of return will be:

- (a) equal to $y\%$ (b) lesser than $y\%$ (c) greater than $y\%$ (d) equal to $xy\%$

(xv) Assertion (A) : 7, (-14), 28, (-56), 112 are in G.P.

Reason (R) : The terms of a G.P. cannot have both positive and negative numbers.

- (a) A is true, R is false (b) both A and R are true
(c) A is false, R is true (d) both A and R are false

Question 2

a) Binay deposits ₹ 500 per month in a recurring deposit account in SBI for 2 years at 6% p.a. At the same time Rahul deposits ₹ 400 per month in a recurring deposit account in Bandhan bank at 8% p.a. for the same period. Who earned more interest and by how much? [4]

b) If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, using k-method, find the value of :

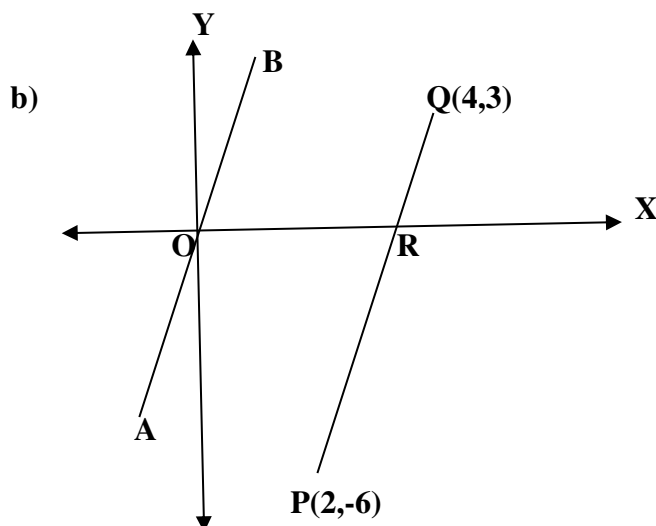
$$\frac{ax-by}{(a+b)(x-y)} + \frac{by-cz}{(b+c)(y-z)} + \frac{cz-ax}{(c+a)(z-x)} \quad [4]$$

c) 11, 22, 44, 88, ... are in G.P.

Find the n^{th} term of this G.P. in terms of n. How many terms of the above G.P. will give the sum 341? [4]

Question 3

a) Solve the following equation correct to two decimal places : $x^2 - 10x + 6 = 0$ [4]



The straight line PQ divides x-axis at R. Find:

(i) Equation of the line PQ.

(ii) The value of k, if PQ passes through $(k-1, k+4)$

(iii) The ratio PR: QR

(iv) Equation of the line AB, if $AB \parallel PQ$. [4]

c) Use graph paper for this question and take 2 cm = 1 unit along both the axes.

i) Plot A(0, 5), B(2, 3) and C(4, -1).

ii) Reflect B and C in the y-axis and name their images as B' and C'.

iii) Reflect A in the line BB' and name its image as A'

iv) Plot and write the co-ordinates of A'.

iv) Join the points ABA'B' and give the geometrical name of the closed figure so formed. [5]

SECTION B

(Attempt *any four* questions from this section)

Question 4

a) Rishi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 orange, 6 blue, and 7 black fishes. Find the probability that the fish is:

(i) Orange fish (ii) Blue or black fish (iii) not a blue fish [3]

b) If $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$, find the value of $(A-2I)(A-3I)$. I is the unit matrix of order 2×2 . [3]

c) If $(x - a)$ is a factor of $x^3 - ax^2 + x + 2$ then, factorize $x^3 - ax^2 + x + 2$ completely. [4]

Question 5

a) Solve the inequation and represent the solution on the number line .

$$-3 + x \leq \frac{7x}{2} + 2 < 8 + 2x, x \in \mathbb{R}. [3]$$

b) The sum of three numbers in A.P. is 24 and the sum of their cubes is 1584. Find the numbers. [3]

c) Johny invests ₹ 22000 in buying ₹ 100 shares at ₹ 10% premium. The dividend is 12%. Find:
i) the number of shares he buy ii) his yearly dividend iii) the percentage return on his investment. [4]

Question 6

a) If $(x - 1)$ is a factor of the expression $3x^3 + ax^2 + bx + 4$ and when the expression is divided by $(x - 2)$, it leaves the remainder 30. Find the value of a and b. [3]

b) Find the equation of the line perpendicular to the line $3x - 2y = 6$ and passing through the point $(0, 1)$ [3]

c) The following bill shows the GST rate, discount rate and the marked price of articles:

Item	Quantity	Rate per item (₹)	Discount	GST
Refrigerator	1	₹ 25000	10%	18%
Microwave	1	₹ 6000	NIL	12%
Bluetooth speaker	2	₹ 2000	5%	28%

Calculate i) The total amount of GST paid.

ii) The total bill amount paid. [4]

Question 7

a) If the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal, show that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. [3]

b) A model of ship is made to a scale 1 : 400.

i) The length of the model of the ship is 1.5m. Calculate the length of the ship

ii) The area of the deck of the ship is 320000 m². Calculate the area of the model.

iii) The volume of the model is 4.5cm³. Calculate the volume of the ship. [3]

c) If $\frac{1+x+x^2}{1-x+x^2} = \frac{62(1+x)}{63(1-x)}$, using properties of proportion solve for x. [4]

Question 8

a) A box contains 20 discs which are numbered as 2, 5, 8, 11, If one disc is drawn from the box, find the probability that it bears (i) a two digit number, (ii) a perfect square number, (iii) a number divisible by 5. [3]

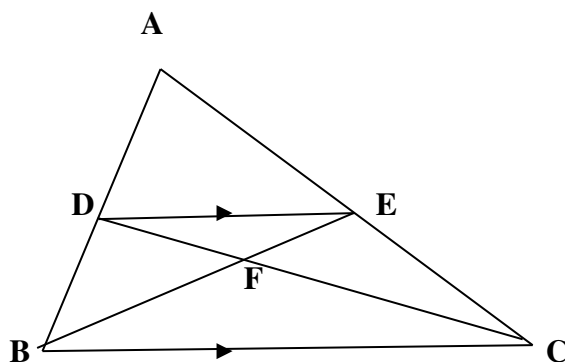
b) Given $\begin{bmatrix} 1 & 1 \\ -3 & 4 \end{bmatrix} \cdot X = \begin{bmatrix} -7 \\ -84 \end{bmatrix}$. Write i) the order of matrix X. ii) the matrix X. [3]

c) In the figure given below, DE || BC and AD:DB = 4:3.

(i) Show that $\triangle ADE \sim \triangle ABC$.

(ii) Find DE:BC.

(iii) Find $\frac{\text{area}(\triangle DEF)}{\text{area}(\triangle DEC)}$. [4]



Question 9

a) Determine the ratio in which the point P(m,6) divides the join of A(-4,3) and B(2,8). Also, find the value of m. Show that the line AB is perpendicular to a line 6x+5y=11. [3]

b) Arnesh invests ₹ 60000 in 8% ₹ 10 shares at ₹ 15. He sells the shares when the price rises to ₹30, and invests the proceeds in 12% ₹ 100 shares at ₹ 125. Calculate:

i) the sale proceeds

ii) the number of ₹ 125 shares he buys

iii) the change in his annual income from dividend. [3]

c) If the marked price of a book is reduced by ₹ 5, a shopkeeper can buy 5 more books for ₹300. Taking the marked price as 'x', create a quadratic equation and hence find the original marked price and the number of books bought. [4]

Question 10

a) The point P(a,b) is first reflected in the origin and then reflected in the x-axis to the point P'(3,-4).

(i) Find the values of a and b.

(ii) P'' is the image of P' when reflected in the origin. Write down the co-ordinates of P''.

(iii) Name the single transformation that maps P to P''. [3]

b)) Find the values of p and q when,

$$A = \begin{bmatrix} p & 0 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 0 & -q \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix} \text{ and } BA = C^2. \quad [3]$$

c) In an A.P. if $S_5 + S_7 = 167$ and $S_{10} = 235$, then find the A.P, where S_n denotes the sum of its first n terms. [4]